

Beginning Group Theory

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Groups and Subgroups

Problem 1. Let x, y, z , and w be elements of a group G .

1. Solve for y , given that $xyz^{-1}w = 1$.

2. Suppose that $xyz = 1$. Does it follow that $yzx = 1$? Does it follow that $yxz = 1$?

Solution. 1. Suppose $xyz^{-1}w = 1$.

$$\begin{aligned}xyz^{-1}w &= 1 \\x^{-1}xyz^{-1}ww^{-1} &= x^{-1}1w^{-1} \\1yz^{-1}1 &= x^{-1}w^{-1} \\yz^{-1}z &= x^{-1}w^{-1}z \\y &= x^{-1}w^{-1}z\end{aligned}$$

2. Suppose that $xyz = 1$.

(a)

$$\begin{aligned}xyz &= 1 \\x^{-1}xyz &= x^{-1}1 \\yz &= x^{-1} \\yzx &= x^{-1}x \\yzx &= 1\end{aligned}$$

$$xyz = 1 \implies yzx = 1$$

(b)

$$\begin{aligned}xyz &= 1 \\x^{-1}xyz &= x^{-1}1 \\yzx^{-1} &= x^{-1}z^{-1} \\yx &= x^{-1}z^{-1}x \\yxz &= x^{-1}z^{-1}xz\end{aligned}$$

In order to get $yxz = 1$, the product $x^{-1}z^{-1}xz$ would need to cancel to 1. Which is only possible if G is commutative or $x = z^{-1}$. Thus $yxz = 1$ doesn't follow from $xyz = 1$. ■

Problem 2. In which of the following cases is H a subgroup of G ?

1. $G = GL_n(\mathbb{C})$ and $H = GL_n(\mathbb{R})$.
2. $G = \mathbb{R}^\times$ and $H = \{1, -1\}$.
3. $G = \mathbb{Z}^+$ and H is the set of positive integers.
4. $G = \mathbb{R}^\times$ and H is the set of positive reals.
5. $G = GL_2(\mathbb{R})$ and H is the set of matrices $\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$ with $a \neq 0$

Solution. 1. $G = GL_n(\mathbb{C})$ and $H = GL_n(\mathbb{R})$. By definition of real and complex numbers $H \subset G$

- (a) **Identity:** $\det(I_n) = 1 \neq 0$, thus I_n is invertible by definition of invertible matrices. Thus $I_n \in GL_n(\mathbb{R})$. Furthermore by definition of Identity matrix, $\forall A \in GL_n(\mathbb{R})$, $AI_n = A = I_nA$. Thus I is the Identity in $GL_n(\mathbb{R})$.
- (b) **Inverse:** $\forall A \in GL_n(\mathbb{R})$, since A is invertible, $\exists A^{-1}$. Furthermore $AA^{-1} = I = A^{-1}A$. Thus by the very definition of invertible matrices, $A^{-1} \in GL_n(\mathbb{R})$.
- (c) **Closure:** $\forall A, B \in GL_n(\mathbb{R})$. By the property of inverses there exists $A^{-1}, B^{-1} \in GL_n(\mathbb{R})$. $AB(B^{-1}A^{-1}) = A(BB^{-1})A^{-1}$ since associativity is inherited. $A(I_n)A^{-1} = AA^{-1} = I_n$. Thus AB is invertible because there exists a $(AB)^{-1} = B^{-1}A^{-1}$. Thus $AB \in GL_n(\mathbb{R})$

H is a subgroup of G .

2. $G = \mathbb{R}^\times$ and $H = \{1, -1\}$. $H \subset G$ since $1, -1 \in G$.

- (a) **Identity:** 1 is the Identity of H .
- (b) **Inverse:** Inverse of 1 is 1. Inverse of -1 is -1. Thus for all $a \in H$, $a^{-1} \in H$
- (c) **Closure:** $1 \times 1 = 1 \in H$, $1 \times -1 = -1 \in H$, $-1 \times 1 = -1 \in H$, $-1 \times -1 = 1 \in H$. Thus H is closed under multiplication

3. $G = \mathbb{Z}^+$ and H is the set of positive integers. For a $0 \notin H$. 0 is the additive Identity of integers and thus 0 must be in H for H to have a identity. Furthermore, the identity in a group is unique. Thus there cannot exist a identity in H which is not 0. Thus H is not a subgroup.

4. $G = \mathbb{R}^\times$ and H is the set of positive reals.

- (a) **Identity:** $1 \in H$. $\forall a \in H$, $a * 1 = a$
- (b) **Inverse:** $\forall a \in H$, since H is the set of positive reals then $0 \notin H$ and thus $a \neq 0$. $\frac{1}{a} > 0$, thus $\frac{1}{a} \in H$ and $a \times \frac{1}{a} = 1 = \frac{1}{a} \times a$. Thus for all $a \in H$, $\exists a \frac{1}{a} \in H$ and a has a inverse.
- (c) **Closure:** For all $a, b \in H$, by definition of multiplication on real numbers $a \times b > 0$ thus $a \times b \in H$

5. $G = GL_2(\mathbb{R})$ and H is the set of matrices $\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$ with $a \neq 0$. H is not even a subset. ■

Cyclic Groups

Problem 1. Let a and b be elements of a group G . Assume that a has order 7 and that $a^3b = ba^3$. Prove that $ab = ba$.

Proof. Let a and b be elements of a group G . Suppose that a has order 7 (ie. $a^7 = I$) and that $a^3b = ba^3$. Then if we left and right multiply by a^3 , thus $a^6ba^3 = a^3ba^6$. Since $a^3b = ba^3$, $a^6ba^3 = a^6a^3b = a^2b$ and $a^3ba^6 = ba^2$, thus $a^2b = ba^2$. Left and right multiply by a^2 , then $a^4ba^2 = a^2ba^4$. Since $a^2b = ba^2$, then $a^4ba^2 = a^6b$ and $a^2ba^4 = ba^6$, thus $a^6b = ba^6$. Left and right multiply by a^6 , thus $a^5ba^6 = a^6ba^5$ because $a^6 * a^6 = a^{12} = a^5$ because a has order 7. Again since $a^6b = ba^6$, $a^5ba^6 = a^11b = a^4b$ and $a^6ba^5 = ba^4$, thus $a^4b = ba^4$. Left and right multiply by a^4 , thus $a^1ba^4 = a^4ba^1$, $a^4 * a^4 = a^8 = a^1$ because a has order 7. Since $a^4b = ba^4$, then $a^1ba^4 = a^5b$ and $a^4ba^1 = ba^5$, thus $a^5b = ba^5$. Again left and right multiply by a^5 , thus $a^3ba^5 = a^5ba^3$ ($a^5 * a^5 = a^{10} = a^3$ because a has order 7). Finally because $a^5b = ba^5$, $a^3ba^5 = a^8b = ab$ and $a^5ba^3 = ba^8 = ba$ because a has order 7. Thus $ab = ba$. $\square$

Claim 2a. The n th roots of unity form a cyclic subgroup of $\mathbb{C}^\times$ of order n .

Proof. Let H be the set of the n th roots of unity. By definition of the n th roots of unity, the k th root of unity is of the form $e^{\frac{i2\pi k}{n}}$. So

$$H = \{e^{\frac{i2\pi k}{n}} : k \in [0, \dots, n-1]\}$$

All elements of H are complex numbers so $H \subset G$.

1. **Identity:** $1 = e^{0i} \in H$. Thus H has a identity
2. **Inverse:** $\forall a \in H$, by definition $a = e^{\frac{i2\pi k}{n}}$ for some $k \in [0, \dots, n-1]$. Let $m = n - k$, $\exists b \in H$ s.t $b = e^{\frac{i2\pi m}{n}}$. $ab = e^{\frac{i2\pi(m+k)}{n}} = e^{\frac{i2\pi n}{n}} = e^{i2\pi} = e^{0i} = 1$. Similarly, $ba = e^{\frac{i2\pi(k+m)}{n}} = e^{\frac{i2\pi n}{n}} = e^{i2\pi} = e^{0i} = 1$. Thus $ab = 1 = ba$. Thus every element has a inverse.
3. **Closure:** For all $a, b \in H$. By definition $a = e^{\frac{i2\pi k}{n}}$ and $b = e^{\frac{i2\pi m}{n}}$ where $k, m \in [0, \dots, n-1]$. $ab = e^{\frac{i2\pi k}{n}} e^{\frac{i2\pi m}{n}} = e^{\frac{i2\pi(m+k)}{n}}$. If $m+k \in [0, \dots, n-1]$, then by definition $ab \in H$. Else, if $m+k > 0$, then $e^{\frac{i2\pi(m+k)}{n}} = e^{\frac{i2\pi(n+t)}{n}}$ for some $t < n$. Then $e^{\frac{i2\pi(n+t)}{n}} = e^{\frac{i2\pi n}{n}} e^{\frac{i2\pi t}{n}} = e^{i2\pi} e^{\frac{i2\pi t}{n}} = e^{\frac{i2\pi t}{n}} \in H$. Thus $ab \in H$

Thus H is a subgroup of G .

Let $\omega = e^{\frac{i2\pi}{n}}$. $\forall a \in H$, by definition $a = e^{\frac{i2\pi k}{n}}$ for some $k \in [0, \dots, n-1]$. Then $\omega^k = \prod_i^k e^{\frac{i2\pi}{n}} = e^{\frac{i2\pi(k*1)}{n}} = e^{\frac{i2\pi k}{n}} = a$. Thus every element of H can be written as a power of ω . Thus

$$H = \{\omega^k : k \in [0, \dots, n-1]\}$$

Thus H is a cyclic subgroup of $\mathbb{C}^\times$ $\square$

Problem 2b. Determine the product of all the n th roots of unity.

Solution.

$$\begin{aligned}
\prod_{k=0}^{n-1} e^{\frac{i2\pi k}{n}} &= e^{\frac{i2\pi \sum_{k=0}^{n-1} k}{n}} \\
&= e^{\frac{i2\pi \frac{n(n-1)}{2}}{n}} \\
&= e^{i\pi(n-1)} \\
&= (-1)^n
\end{aligned}$$

■

Claim 3. *Let a and b be elements of a group G . Prove that ab and ba have the same order.*

Proof. Suppose ab has order k . Then by definition $(ab)^k = 1$. Left multiply by a , then $(ab)^k a = ((ab) \dots (ab))a$ (ab is multiplied k times with itself). By property of associativity, $((ab) \dots (ab))a = a((ba) \dots (ba)) = a(ba)^k$ (Just changed the where the parenthesis was). Since $(ab)^k = 1$, then $(ab)^k a = 1a = a$. Thus $a = a(ba)^k$. Thus by definition of identity, $(ba)^k$ must be the identity. Thus $(ba)^k = 1$ and ba has the same order as ab .

□

Problem 4. *Describe all groups G that contain no proper subgroup.*

Solution. All groups G that contain no proper subgroups are the cyclic groups with prime order. This is because by Lagrange's Theorem the order of the subgroup divides the order of the group. If G is a subgroup with no proper subgroup then the orders of its subgroups are 1 or $n = |G|$ by definition. Thus the order of G can only be divided by n or 1. Thus order of G must be prime.

■

Claim 5. *Prove that every subgroup of a cyclic group is cyclic. Do this by working with exponents, and use the description of the subgroups of $\mathbb{Z}^+$.*

Proof. Suppose G is a cyclic group. Then by definition of a cyclic group, there exists a $g \in G$ s.t $G = \langle g \rangle = \{g^k : k \in \mathbb{Z}\}$. Let the identity element of G be e and let H be a subgroup of G .

If $H = \{e\}$, then the subgroup is cyclic because $e^m = e$ for any $m \in \mathbb{Z}$.

If $H \neq \{e\}$, assume the contrary, that H is not cyclic. Let $a, b \in H$. Because H is a subgroup of G , a and b are also in G and thus can be represented as g^m for $m \in \mathbb{Z}$. Let p be the smallest positive integer such that $g^p \in H$, let $a = g^p$. Given that H is not cyclic there must be a element $b = g^q$ s.t $b = g^q \neq g^{kp}$ for a $k \in \mathbb{Z}$. Thus q is not a integer multiple of p and can be written as $q = kp + r$ for $k \in \mathbb{Z}$ and $r \in \mathbb{Z}$ where $0 < r < p$. Thus $b = g^q = g^{kp+r}$. Because $a = g^p \in H$, then by the property of closure $a^k = (g^p)^m = g^{mp} \in H$. Furthermore, because each element must have a inverse in a group, g^{-mp} must also be in the group. Furthermore because $b = g^{mp+r}$ and g^{-mp} are in the H , by the property of closure $b * g^{-mp} = g^r$ must also be in H . But we asserted because q is not an integer multiple p , r must be a nonzero positive integer less than p . But since we asserted that p is the smallest positive integer s.t $g^p \in H$ but since g^r is also in H and $r < p$, a contradiction forms and H is cyclic.

□

Problem 6a. Let G be a cyclic group of order 6. How many of its elements generate G ? Answer the same question for cyclic groups of orders 5 and 8.

Solution. 1. Let $G = \langle x \rangle$, be the cyclic group of order 6. 2 of its elements generate G , the elements are x^1, x^5 .

2. Let $G = \langle x \rangle$, be the cyclic group of order 5. All of its elements can generate G .

3. Let $G = \langle x \rangle$, be the cyclic group of order 8. Only 4 elements can generate G , the elements are x, x^3, x^5, x^7 . ■

Problem 6b. Describe the number of elements that generate a cyclic group of arbitrary order n .

Solution. Let k be coprime to n . Then by Bezouts Theorem, $\exists a, b \in \mathbb{Z}$ s.t $ak + bn = 1$. For all $m \in \mathbb{Z}$

$$\begin{aligned} a^m &= a^{m(1)} \\ &= a^{m(ak+bn)} = a^{mak+mbn} = a^{mak} a^{mbn} = a^{mak} e = a^{mak} = (a^k)^m a \end{aligned}$$

Thus a^k generates the cyclic group.

Let the cyclic group be generated by a^k . Then there exists a u s.t $(a^k)^m = a$. Thus $mk \equiv 1 \pmod{n}$. By Bezouts Theorem, $\exists x, y \in \mathbb{Z}$ s.t $1 = xk + yn$. Thus $\gcd(k, n) = 1$ and k is coprime to n .

The number of elements that generate a cyclic group of arbitrary order n is equal to the number of coprime integers to n or $\phi(n)$ where ϕ is Euler's totient function. ■